4.3 Real Options: don't make commitments before you have to

In the mid-2000s, Chris Matts identified a fundamental principle underlying many Agile practices: putting off decisions until the "last responsible moment," an idea that comes from lean software development. He called this principle *Real Options*. Understanding this principle changes the way you think about many Agile practices and opens the door to a few new ones.

In finance, an *option* gives you the possibility, but not the obligation, to purchase a product sometime in the future at today's price. For example, imagine that there's a high probability that you'll need to purchase a large quantity of steel in the next three months, and that the price of steel is currently on the rise. You don't want to buy the steel now, because you aren't completely sure that you'll need it; you expect to know for sure sometime in the next two months. But if you wait another few months, the price of steel might have gone up, which means that you'll lose money. To get out of this conundrum, you can buy an option to purchase the steel sometime within the next three months, at today's price. If the price of steel goes up, you can still buy at today's price. And if the price goes down, or if you don't need the steel, you can choose not to use the option. You need to pay for this option, but it only costs a fraction of the total price of the steel: it's worthwhile because it allows you to not commit yourself to buying the steel until you're sure you need it.

This principle also applies in day-to-day life. When you buy a plane ticket, you're actually buying an option to travel: the ticket places you under no obligation to travel. But the price you pay for this option varies. Imagine your favorite airline is offering tickets for only $600 to go from Sydney to Wellington, but these cheaper tickets are nonrefundable if you decide not to travel. You're not sure that you'll be able to make the trip, so you opt for a more expensive $800 ticket, which has a $25 cancellation fee.

Let's look at the math here. The option to cancel the flight costs you an extra $200. If you're fairly likely to travel, this might be a lot to pay for an option you're unlikely to use, so you might prefer the cheaper ticket. But if you think that there's a 50% chance that you won't be able to fly, you may well be happy to pay the extra $200. If you cancel, you'll only lose $225 (the extra $200, plus the $25 cancellation fee), whereas if you cancel after opting for the cheaper flight, you'll lose $600.

Real Options is an application of these principles to software development invented by Chris Matts (see figure 4.17).[4] Chris summarizes the principles of Real Options in three simple points:

- Options have value.
- Options expire.
- Never commit early unless you know why.

[4] Chris Matts and Olav Maassen, "'Real Options' Underlie Agile Practices," InfoQ (2007), http://www.infoq.com/articles/real-options-enhance-agility.